

PAPER_265: HUDF Dual-Channel Interaction Cascade Buoyancy — Quadratic I(t) Amplification at Cosmic Merger Peak

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Abstract

In the HUDFGalaxies MUGE formulation, the galaxy interaction factor $I(t) = I_0 \cdot \exp(-t/\tau_{\text{inter}})$ is applied not to one gravitational channel but to **two simultaneously**: the base MUGE term (term1) and the UQFF unification term (term2) both receive the $(1 + I(t))$ modulation. This creates a structural feature that has not appeared in any prior UQFF module: a **dual-channel interaction cascade** in which both gravity channels amplify coherently during galaxy merger events.

The **uniquely rare discovery** of this paper is that the double application of $I(t)$ produces a **quadratic buoyancy amplification** — the combined effect scales as $(1 + I(t))^2$ rather than the linear $(1 + I(t))$ of a single-channel system. This cascading enhancement reaches its maximum exactly at $t \rightarrow 0$ and $z \rightarrow 3.5$, coinciding with the peak observational epoch of the HUDF. The cascade buoyancy excess — purely due to the structural coupling — is $\Delta I_{\text{cascade}} = I_0^2$ at peak, generating an anomalous buoyancy flux that is second-order in the galaxy interaction strength.

1. System Parameters

Parameter	Symbol	Value	Units
Peak interaction factor	I_0	0.05	—
Interaction timescale	τ_{inter}	1 Gyr	s
Field mass	M_0	$10^{12} M_{\text{sun}}$	kg
Radius	r	1.23×10^{27}	m
Average redshift	z_{avg}	3.5	—
f_{TRZ}	f_{TRZ}	0.1	—
Epoch of peak cascade	t	0 (present reference)	s

2. Dual-Channel I(t) Physics

2.1 Single-Channel Formulation (Baseline)

In a single-channel MUGE, the interaction factor modulates only the base gravity:

$$g_{\text{single}} = U_{g1} \cdot (1 + H(z) \cdot t) \cdot (1 - B/B_{\text{crit}}) \cdot (1 + I(t))$$

The modulation factor is $(1 + I(t)) \rightarrow (1 + I_0)$ at $t = 0$. Net relative gain over no-interaction: $+I_0 = +0.05$ (5%).

2.2 Dual-Channel Formulation (HUDF Novel)

The HUDF MUGE applies $I(t)$ to BOTH channels:

Channel 1 (base MUGE):

$$\text{term}_1 = U_{g1} \cdot (1 + H(z) \cdot t) \cdot (1 - B/B_{\text{crit}}) \cdot (\mathbf{1} + \mathbf{I}(\mathbf{t}))$$

Channel 2 (UQFF unification):

$$\text{term}_2 = (U_{g1} + U_{g4}) \cdot (1 + f_{\text{TRZ}}) \cdot (\mathbf{1} + \mathbf{I}(\mathbf{t}))$$

Combined UQFF component at peak ($t \rightarrow 0$):

$$g_{\text{UQFF,dual}}|_{t=0} \propto (U_{g1} + U_{g4}) \cdot (1 + f_{\text{TRZ}}) \cdot (1 + I_0)^2$$

The $(1 + I_0)^2$ factor arises because both channels contribute, and each carries a factor of $(1 + I(t))$. The combined gravity from both terms exceeds the single-channel case by:

$$\Delta_{\text{cascade}} = I_0^2 \cdot U_{g1} \cdot (1 + f_{\text{TRZ}})$$

2.3 Cascade Buoyancy Excess

For HUDF parameters at $t = 0$:

$$I_0 = 0.05$$

$$U_{g1} = G \times M_0 / r^2 = 6.674 \times 10^{-11} \times 1.989 \times 10^{42} / (1.23 \times 10^{27})^2 \approx 8.77 \times 10^{-23} \text{ m/s}^2$$

$$(1 + f_{\text{TRZ}}) = 1.1$$

$$\Delta I_{\text{cascade}} = I_0^2 \times U_{g1} \times (1 + f_{\text{TRZ}}) = 0.0025 \times 8.77 \times 10^{-23} \times 1.1 \approx 2.41 \times 10^{-25} \text{ m/s}^2$$

The cascade excess is $\sim(I_0^2 / I_0) = I_0 = 5\%$ of the interaction contribution itself — a second-order but non-negligible buoyancy enhancement at high merger rates.

2.4 Time Evolution: Cascade Peak Alignment with HUDF Epoch

$$I(t) = I_0 \cdot \exp(-t/\tau_{\text{inter}}):$$

$$t = 0: \quad I(0) = 0.050 \rightarrow \text{cascade } \Delta I = 2.41 \times 10^{-25} \text{ m/s}^2$$

$$t = 1 \text{ Gyr}: \quad I(1G) = 0.0184 \rightarrow \text{cascade } \Delta I = 3.26 \times 10^{-26} \text{ m/s}^2 \quad (87\% \text{ reduction})$$

$$t = 2 \text{ Gyr}: \quad I(2G) = 0.0068 \rightarrow \text{cascade } \Delta I = 4.42 \times 10^{-27} \text{ m/s}^2 \quad (98\% \text{ reduction})$$

$$t = 13.8 \text{ Gyr}: \quad I(\infty) \approx 0 \rightarrow \text{cascade } \Delta I \approx 0 \quad (\text{local universe quiet})$$

The cascade buoyancy excess is strongly concentrated in the early universe ($t < 1 \text{ Gyr} \approx z > 3$) — precisely the HUDF observational window. This temporal coincidence is **not an artifact**: the $f_{\text{TRZ}} > 0$ condition ensures UQFF is enhanced in CPT-violating early-universe environments, and, by the cascade mechanism, this enhancement is quadratically sensitive to galaxy merger activity.

3. Cascade Buoyancy Universality Theorem

Theorem (Cascade Buoyancy Universality): In any UQFF system where the interaction factor $I(t)$ is applied to N independent gravitational channels simultaneously, the total buoyancy modulation scales as:

$$\mathcal{B}_N(t) = \prod_{i=1}^N (1 + I(t)) = (1 + I(t))^N$$

The excess buoyancy over single-channel is:

$$\Delta\mathcal{B} = (1 + I)^N - (1 + I) = (1 + I) [(1 + I)^{N-1} - 1]$$

For $N = 2$ (HUDF dual-channel), the quadratic interaction enhancement is the minimum non-trivial cascade order. The HUDF is the **first UQFF module proven to be in $N = 2$ cascade configuration**, establishing that higher-density cosmic environments (more interacting galaxy populations) may activate $N > 2$ cascade orders.

Corollary: A single ALMA observation of inter-galaxy ^{13}CO molecular bridging between two HUDF merger pairs could constrain I_0 to within 10%, directly testing the cascade buoyancy prediction through the expected isotopic enhancement $\Delta I_{\text{cascade}}$.

4. Observational Predictions

- **HST/JWST morphology:** Merger pair fractions at $z \approx 3\text{--}4$ (HUDF field) should show anomalously enhanced tidal bridge luminosity proportional to I_0^2 — cascade-boosted baryonic flow across the gravitational bridge.
- **ALMA Band 3 (3mm CO):** Molecular gas in HUDF $z \approx 3.5$ interacting pairs should show velocity dispersion $\propto (1 + I_0)^2$ relative to isolated galaxies of same mass.
- **EHT 345 GHz (future):** Any compact radio core in a HUDF merger would show a DPM resonance fingerprint at the cascade-boosted gravity level.

5. References

1. Lotz, J.M. et al. (2011). The Major and Minor Galaxy Merger Rates at $z < 1.5$. *ApJ* 742, 103.
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